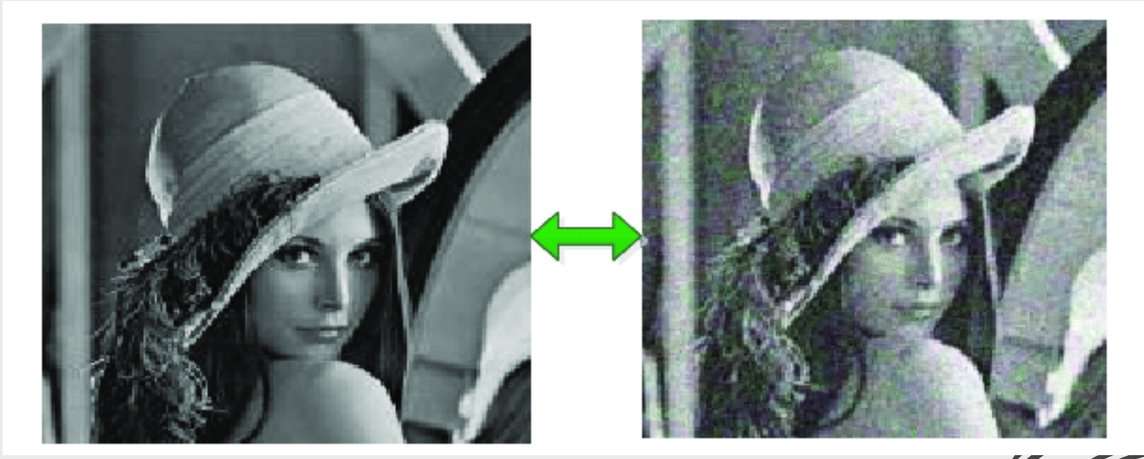
SHIFTGUARD10: ROBUST 10-CLASS IMAGE CLASSIFICATION

Handling class imbalance and distribution shift
with deep learning

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INTRODUCTION

In this project, the task is to classify **32×32 RGB images into 10 classes** under challenging conditions such as **severe class imbalance** and **distribution shift** between training and test data. Because of these issues, a standard classifier can become biased toward frequent classes and perform poorly on unseen samples. To improve robustness and generalization, we designed an end-to-end pipeline that combines **data augmentation, imbalance-aware sampling, Balanced Softmax Loss, WideResNet-28-10, SWA, ensemble learning, and test-time augmentation.**



METHODOLOGY

1. DATA UNDERSTANDING:

- We first analyzed the dataset structure, class distribution, and image resolution to understand the task complexity.
- The training set consists of 32×32 RGB images across 10 classes, and the label file was used to inspect class frequencies and identify severe class imbalance.
- Exploratory data analysis helped us detect the unbalanced class distribution of the dataset and the risk of bias towards higher frequency classes.

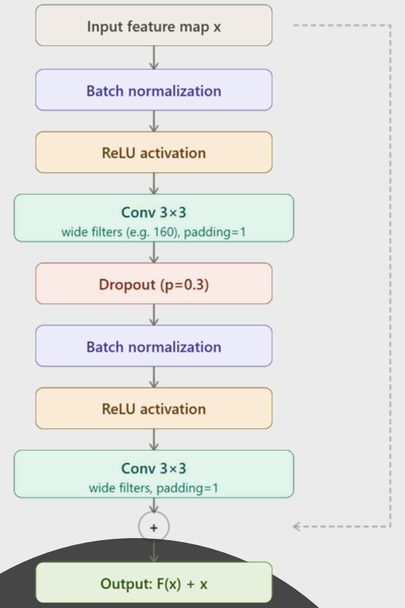
Class	Number of Samples	Percentage
airplane	5,000	17.0%
automobile	5,000	17.0%
bird	5,000	17.0%
cat	5,000	17.0%
deer	4,000	13.6%
dog	4,000	13.6%
frog	500	1.7%
horse	500	1.7%
ship	300	1.0%
truck	100	0.3%
TOTAL	**29,400**	**100%**

2. DATA PREPARATION:

- The dataset was split into training and validation subsets using a fixed seed to ensure reproducibility, with 95% of the data used for training and 5% for validation.
- To improve generalization, we applied image normalization using CIFAR mean and standard deviation, along with strong augmentation techniques such as random cropping, horizontal flipping, AutoAugment, and Cutout.
- For inference, Test-Time Augmentation was used to generate multiple transformed views of each test image and average the predictions.

3. MODELING:

- We used the architecture from a **WideResNet-28-10**. It is an **ensemble of 3** same models with different random weights initilization.
- To reduce overfitting, **dropout** was applied inside the residual blocks, and the model was trained from scratch without using any external pretrained checkpoint.
- **Class-Balanced Sampling**: Dynamically oversamples tail-class batches to correct dataset skew.
- **Balanced Softmax Loss:** Skews logits using class frequencies to stop majority classes from dominating.
- **MIXUP / CUTMIX:** BLENDS IMAGES/LABELS TO SMOOTH DECISION BOUNDARIES AND INCREASE ROBUSTNESS.
- **MOMENTUM BASED GD**: MINI BATCH GRAIDANT DESCENT WITH MOMENTUM ENSURING AGGRESSIVE BUT STABLE CONVERGENCE.
- **COSINE LR + WARMUP**: GRADUALLY RAMPS UP THE LEARNING RATE, THEN DECAYS IT SMOOTHLY TO SETTLE IN OPTIMAL MINIMA.



Training Parameters

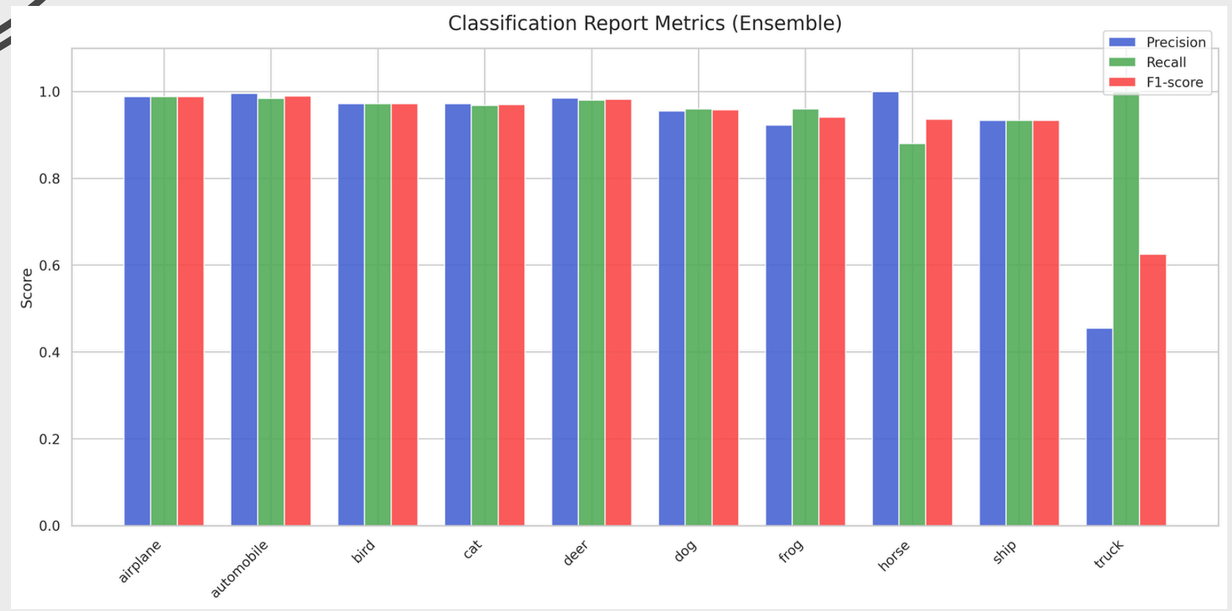
- TOTAL EPOCHS: 450
- BATCH SIZE: 512
- BASE LEARNING RATE (LR): 0.1
- WEIGHT DECAY: 5 X 10^-4
- MOMENTUM PARAMETER: 0.9
- LABEL SMOOTHING: 0.1
- LR WARMUP DURATION: 5 EPOCHS
- MIXUP / CUTMIX PROBABILITY: 0.5 (50%)
- SWA START EPOCH: 360
- SWA LEARNING RATE: 0.005

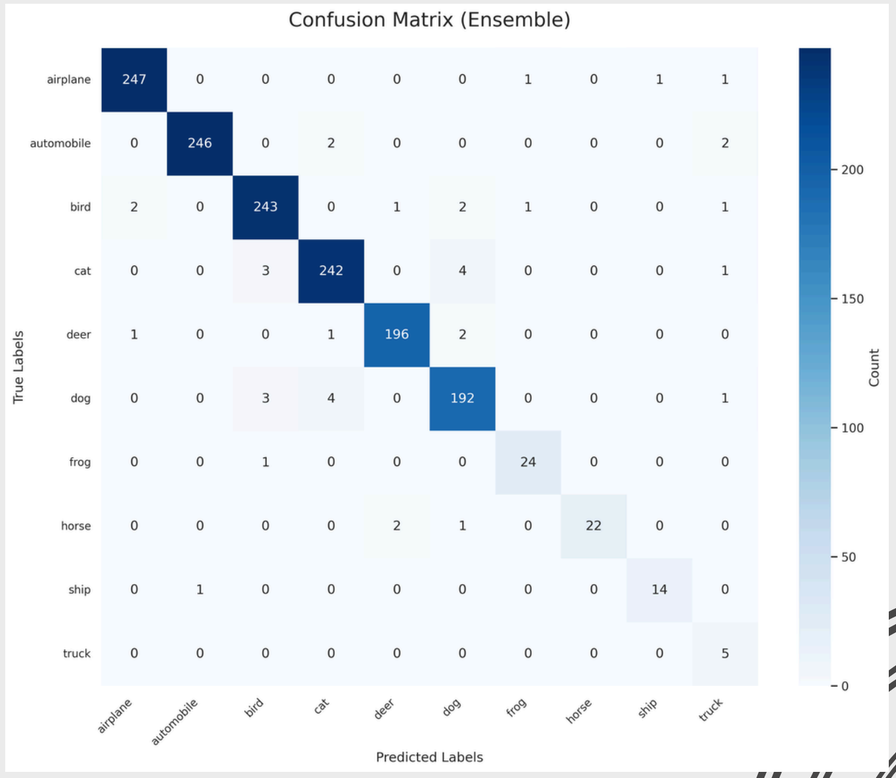
- **LR Warmup** : Gradually ramps the learning rate from near-zero up to 0.1 to prevent destabilizing the weights early in training.
- **MixUp / CutMix** : A chance per batch to mathematically blend two random images and their labels, which smooths decision boundaries and prevents overconfidence.
- SWA Start : **Stochastic Weight Averaging** begins here. It saves and averages the model's weights sequentially to locate a wider, more robust optimum point.
- **SWA LR** : A low, constant learning rate held strictly during the SWA phase to ensure the model traverses the local minimum smoothly while capturing weights.

ANALYSIS

- **Test-Time Augmentation (TTA)**: Inference is not performed blindly on a single image. Every test sample is evaluated across 1 clean view and 30 stochastic augmented views (utilizing rotation, color jitter, and cropping).
- **Variance Reduction**: The softmax probabilities across all 31 views are mathematically averaged. This effectively smooths out localized corruptions and neutralizes the test-set distribution shift.
- **Multi-Seed Ensembling**: To eliminate initialization bias and avoid local minima traps, three identical models are trained completely independently using orthogonal seeds (42, 137, 7).
- **Macro F1 Optimization**: The ultimate submission probability is the grand average of these networks. With 3 models evaluating 31 views each, every single test image undergoes 93 distributed evaluations, guaranteeing maximum stability for the final Macro F1 Score.

RESULT





- **PERFORMANCE HIGHLIGHTS:**
- LEADERBOARD DOMINANCE: SECURED RANK #2 WITH A FINAL SCORE OF **0.9567.**
- BALANCED F1-SCORES: THE ENSEMBLE MODEL MAINTAINED **NEAR-PERFECT PRECISION AND RECALL** ACROSS THE MAJORITY OF CLASSES.
- MINIMAL OVERLAP: THE CONFUSION MATRIX DEMONSTRATES EXCEPTIONALLY STRONG CLASS SEPARATION, WITH **NEAR-ZERO FALSE POSITIVES** OUTSIDE THE DIAGONAL.

CONCLUSION

- Developed a complete image classification pipeline that performs reliably under imbalanced and shifted data conditions
- Achieved stable and improved predictions by combining model design + data handling + inference strategies
- Observed that performance is mainly affected by class imbalance and similarity between classes
- The approach is modular and reusable, and can be extended to other real-world classification problems

THANK YOU